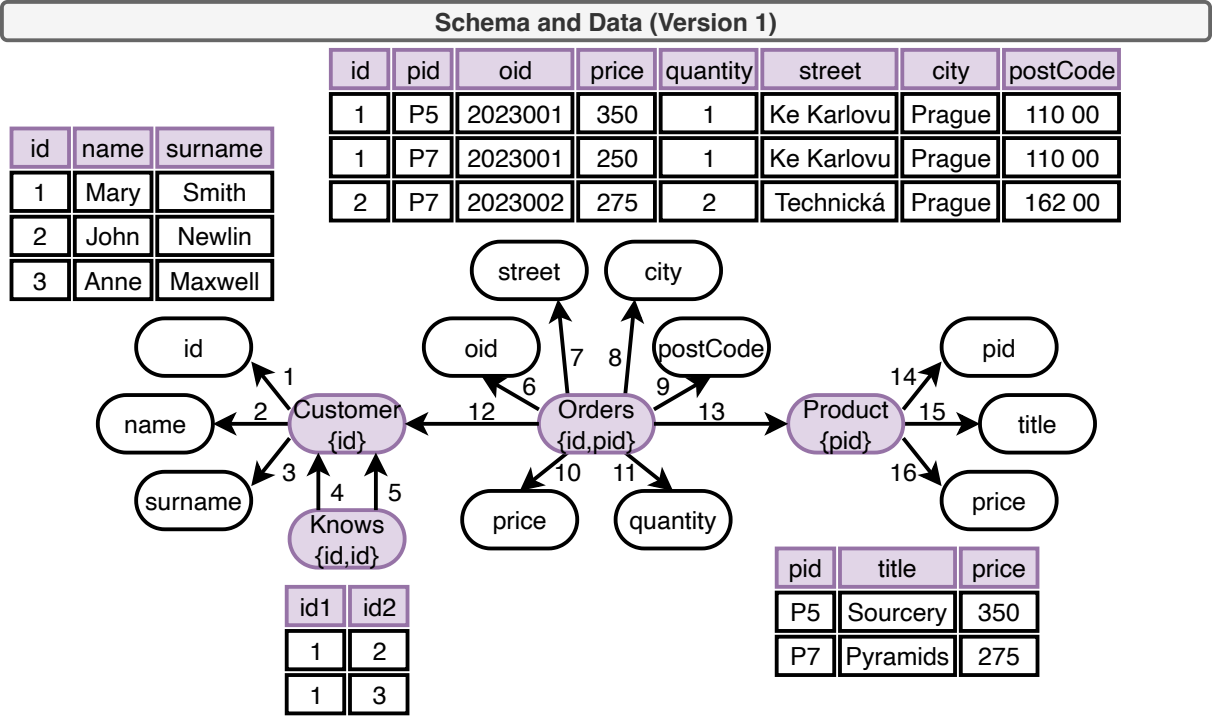




Query 1 (Version 1)

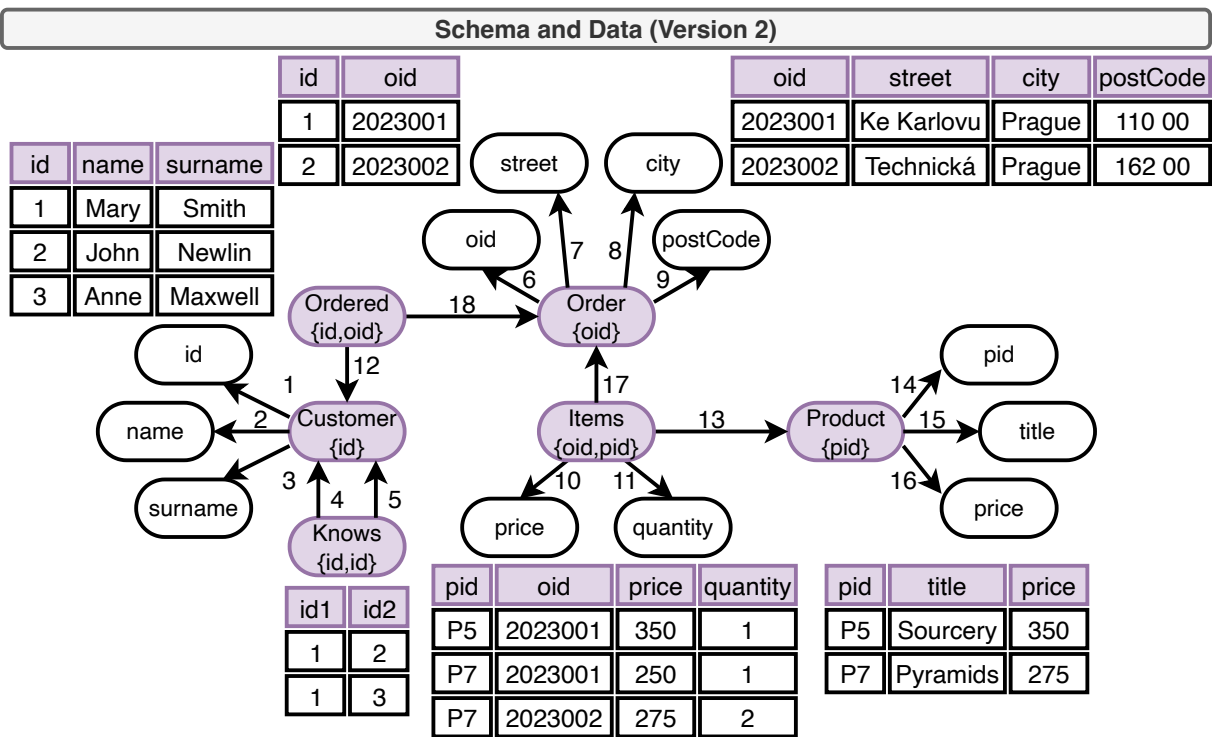
```
SELECT {
  _:knows friend ?friendA ;
  friend ?friendB ;
  ?friendA fName ?friendAFName ;
  IName ?friendALName .
  ?friendB fName ?friendBFName ;
  IName ?friendBLName .
} WHERE {
  ?friendA -4/5 ?friendB .
  ?friendA 2 ?friendAFName ;
  3 ?friendALName .
  ?friendB 2 ?friendBFName ;
  3 ?friendBLName .
} FILTER(?friendA < ?friendB)
```

SELECT c1.name, c1.surname, c2.name, c2.surname FROM Knows k JOIN Customer c1 ON k.id1 = c1.id JOIN Customer c2 ON k.id2 = c2.id WHERE c1.id < c2.id

Query 2 (Version 1)

```
SELECT {
  SUM(?price) AS ?totalPrice
} WHERE {
  ?orders 12 ?customer ;
  10 ?price .
  ?customer 2 "Anna" .
} ORDER BY (?totalPrice) DESC LIMIT 1
```

SELECT SUM(o.price) AS totalPrice FROM Customer c JOIN Orders o ON c.id = o.id WHERE c.name = 'Anna' ORDER BY totalPrice LIMIT 1



Query 1 (Version 2)

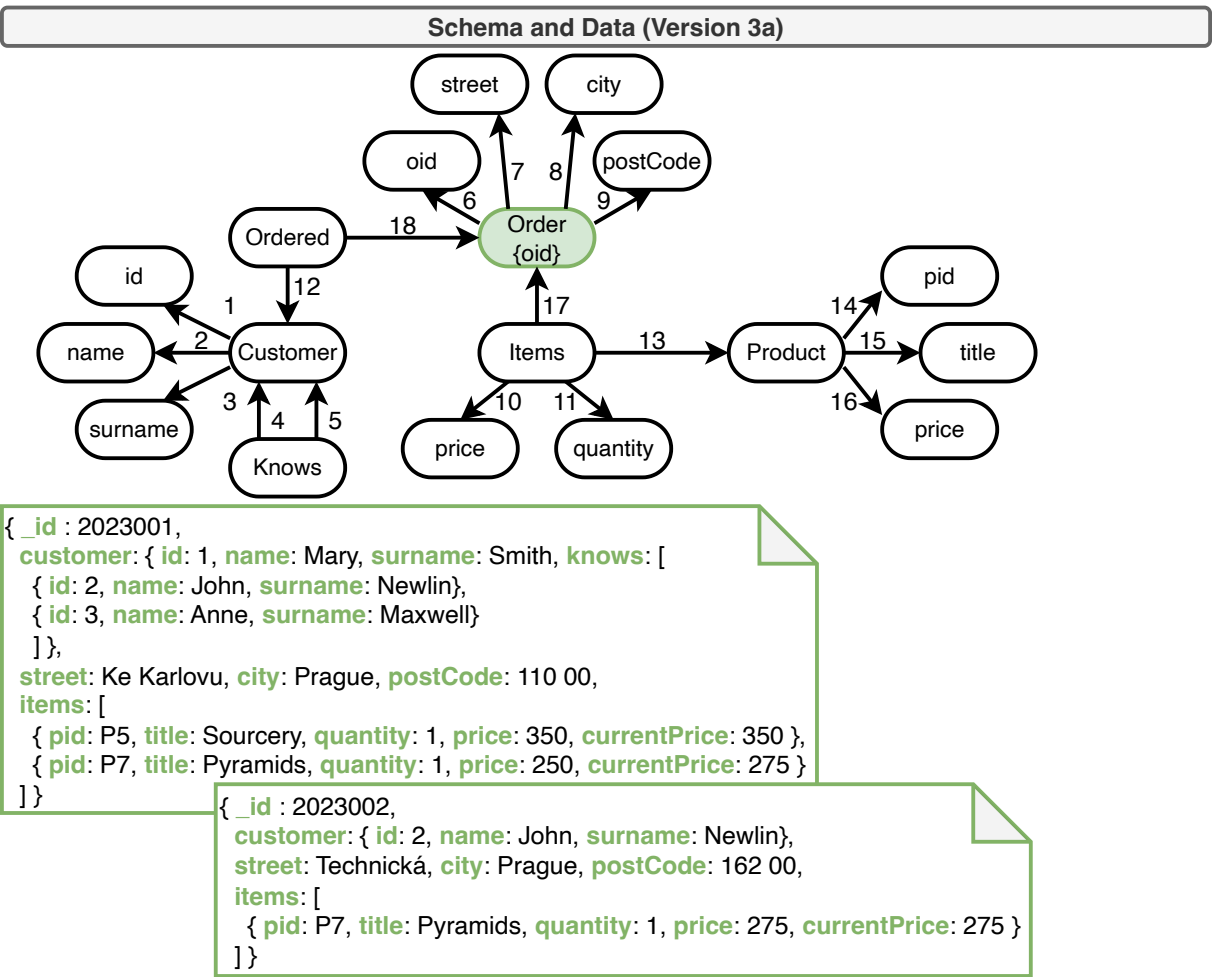
```
SELECT {
  _:knows friend ?friendA ;
  friend ?friendB ;
  ?friendA fName ?friendAFName ;
  IName ?friendALName .
  ?friendB fName ?friendBFName ;
  IName ?friendBLName .
} WHERE {
  ?friendA -4/5 ?friendB .
  ?friendA 2 ?friendAFName ;
  3 ?friendALName .
  ?friendB 2 ?friendBFName ;
  3 ?friendBLName .
} FILTER(?friendA < ?friendB)
```

SELECT c1.name, c1.surname, c2.name, c2.surname FROM Knows k JOIN Customer c1 ON k.id1 = c1.id JOIN Customer c2 ON k.id2 = c2.id WHERE c1.id < c2.id

Query 2 (Version 2)

```
SELECT {
  SUM(?price) AS ?totalPrice
} WHERE {
  ?orders 17/-18/12 ?customer ;
  10 ?price .
  ?customer 2 "Anna" .
} ORDER BY (?totalPrice) DESC LIMIT 1
```

SELECT SUM(o.price) AS totalPrice FROM Customer c JOIN Ordered r ON c.id = r.id JOIN Orders o ON r.oid = o.oid JOIN Items i ON o.oid = i.oid JOIN Product p ON i.pid = p.pid WHERE c.name = 'Anna' ORDER BY totalPrice LIMIT 1



Query 1 (Version 3a)

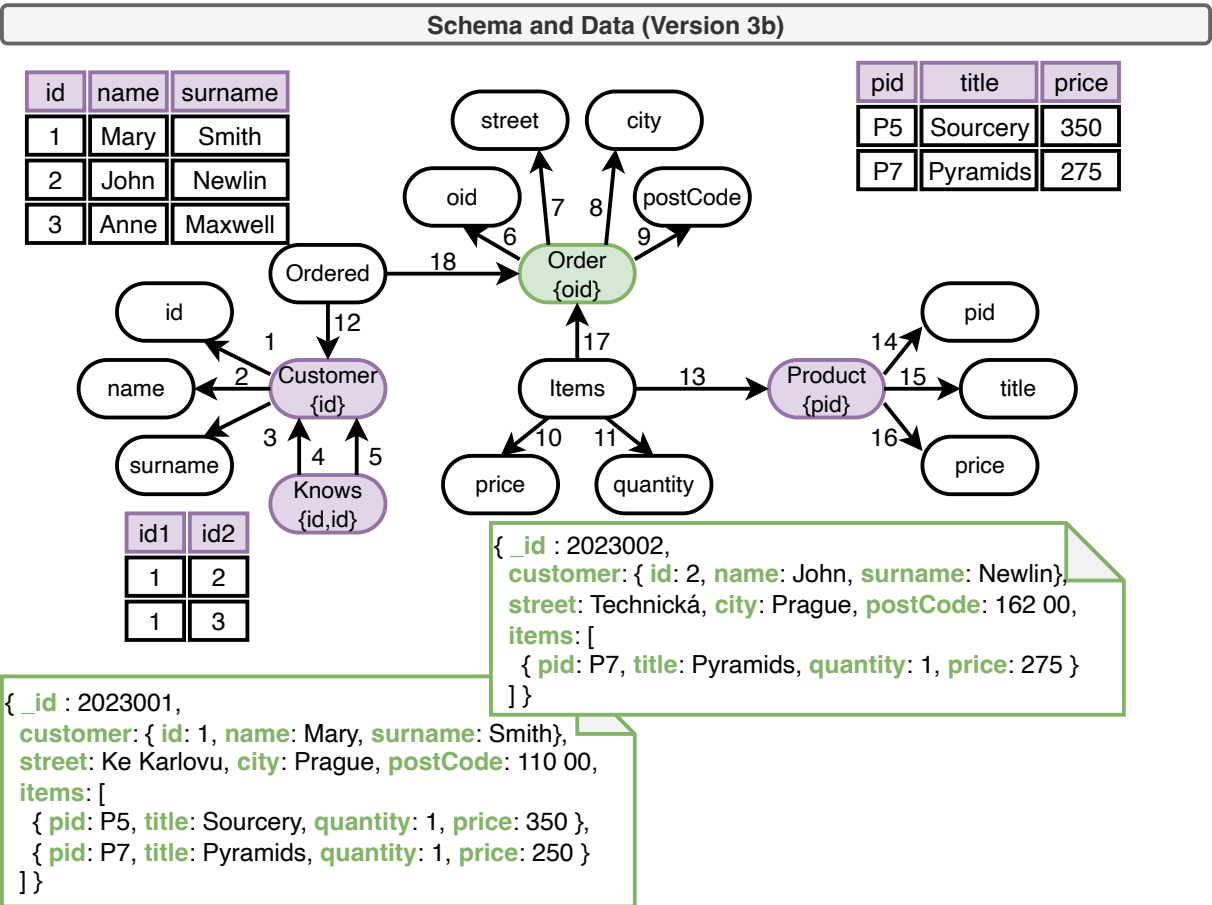
```
SELECT {
  _:knows friend ?friendA ;
  friend ?friendB ;
  ?friendA fName ?friendAFName ;
  IName ?friendALName .
  ?friendB fName ?friendBFName ;
  IName ?friendBLName .
} WHERE {
  ?orders -18/12 friendA .
  ?orders -18/12 friendB .
  ?friendA -4/5 ?friendB .
  ?friendA 2 ?friendAFName ;
  3 ?friendALName .
  ?friendB 2 ?friendBFName ;
  3 ?friendBLName .
} FILTER(?friendA < ?friendB)
```

db.orders.find({ "customer.id": { \$lit: "customer.knows.id" } }, { "_id": 0, "customer.name" : 1, "customer.surname" : 1, "customer.knows.name" : 1, "customer.knows.surname" : 1 })

Query 2 (Version 3a)

```
SELECT {
  SUM(?price) AS ?totalPrice
} WHERE {
  ?orders -18/12 ?customer ;
  -17/10 ?price .
  ?customer 2 "Anna" .
} ORDER BY (?totalPrice) DESC LIMIT 1
```

db.orders.aggregate([{ \$match: { "customer.name" : "Anna" } }, { \$unwind: "\$items" }, { \$group: { _id: "\$_id", totalPrice: { \$sum: "\$items.price" } } }, { \$project: { _id: 0, totalPrice: 1 } }, { \$sort: {totalPrice: -1} }, { \$limit: 1 }]);



Query 1 (Version 3b)

```
SELECT {
  _:knows friend ?friendA ;
  friend ?friendB ;
  ?friendA fName ?friendAFName ;
  IName ?friendALName .
  ?friendB fName ?friendBFName ;
  IName ?friendBLName .
} WHERE {
  ?friendA -4/5 ?friendB .
  ?friendA 2 ?friendAFName ;
  3 ?friendALName .
  ?friendB 2 ?friendBFName ;
  3 ?friendBLName .
} FILTER(?friendA < ?friendB)
```

SELECT c1.name, c1.surname, c2.name, c2.surname FROM Knows k JOIN Customer c1 ON k.id1 = c1.id JOIN Customer c2 ON k.id2 = c2.id WHERE c1.id < c2.id

Query 2 (Version 3b)

```
SELECT {
  SUM(?price) AS ?totalPrice
} WHERE {
  ?orders -18/12 ?customer ;
  -17/10 ?price .
  ?customer 2 "Anna" .
} ORDER BY (?totalPrice) DESC LIMIT 1
```

db.orders.aggregate([{ \$match: { "customer.name" : "Anna" } }, { \$unwind: "\$items" }, { \$group: { _id: "\$_id", totalPrice: { \$sum: "\$items.price" } } }, { \$project: { _id: 0, totalPrice: 1 } }, { \$sort: {totalPrice: -1} }, { \$limit: 1 }]);